Literature Review and Project Plan

Deep BSDE Methods for Multi-Agent Dealer Markets

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Abstract

The Cont–Xiong (2024) multi-agent dealer market defines a coupled system of Hamilton–Jacobi–Bellman equations whose Nash equilibrium is equivalent to the solution of a mean-field forward-backward stochastic differential equation (FBSDE) with jumps. This project studies that FBSDE system computationally, using deep BSDE methods as the numerical tool. The aim is threefold: (i) to approximate the N-player Nash equilibrium via a deep BSDE solver and quantify its approximation error and convergence behaviour, extending the curse-of-dimensionality-free theory of Han, Hu, and Long [2022] to the jump-augmented setting; (ii) to characterise the conditions under which naive deep-learning parameterisations fail to represent the mean-field coupling of the HJB system, and to design parameterisations that do; and (iii) to study the spread gap $|S_{\mathrm{MADDPG}} - S_{\mathrm{Nash}}|$ as a quantitative statistic for deviations of multi-agent reinforcement-learning dynamics from the mean-field Nash equilibrium. A sequential set of basic, intermediate, and advanced goals, with an explicit test harness, structures the 13-week project timeline.

1 Scientific Background

Multi-agent dealer market-making sits at an uncomfortable intersection in the mathematical literature. The N-player stochastic-differential game of Cont and Xiong [2024] models competing market makers posting bid and ask quotes in a continuous-time dealer market; executions arrive as compensated point processes, and Nash equilibrium is characterised by a coupled system of Hamilton–Jacobi–Bellman (HJB) equations. The game admits *in principle* a probabilistic treatment via backward stochastic differential equations (BSDEs) and mean-field games (MFGs), and a numerical treatment via deep-learning methods. *In practice*, no existing framework simultaneously handles all three ingredients that the model requires: Han et al. [2022] give a curse-of-dimensionality-free deep-BSDE method for McKean–Vlasov forward-backward SDEs but without jumps; Wang, Wang, Li, Wang, and Li [2023] extend deep BSDE to jump-diffusion processes but without mean-field coupling; and Cont and Xiong [2024] solve their own game empirically via multi-agent deep deterministic policy gradient (MADDPG) but without the equilibrium grounding or interpretable gradients that a principled method would supply. Combining all three — jump-augmented BSDEs, mean-field coupling, and deep-learning numerics — into a single computational method for a realistic market microstructure model is the gap this project addresses.

Concretely, each dealer controls the bid/ask spread δ_t over their quote, and the execution probability in response to a random order is governed by the softmax-type kernel of Cont and Xiong [2024] (their eq. 58),

$$f_a^i(\delta, s) \;=\; \tfrac{1}{N}\,\sigma(-\delta)\,\frac{e^s}{1+e^{\delta+s}}, \tag{1}$$

where $\sigma(\cdot)$ is the logistic sigmoid and $s = S/\kappa$ is the market-price scale. Nash equilibrium is the fixed point of the coupled Bellman system, and existence follows from a Brouwer fixed-point argument; what is missing is a scalable numerical method that respects both the mean-field limit and the jump-diffusion structure.

2 Related Work

The literature divides naturally into three pillars.

BSDEs and mean-field games. Pardoux and Peng [1990] proved existence and uniqueness of adapted BSDE solutions under Lipschitz hypotheses, providing the bedrock on which Karoui, Peng, and Quenez [1997] built the BSDE formulation of derivative pricing and risk management. Feng, Wang, and Zhao [2018] extended the PDE/FBSDE correspondence to the *weak*-solution setting for quasi-linear PDEs — directly relevant to the Cont–Xiong HJB system, whose coefficients are quasi-linear rather than classical. A BSDE on $[0, T]$ is a pair of adapted processes (Y_t, Z_t) satisfying

$$Y_t \;=\; \xi + \int_t^T f(s, Y_s, Z_s)\, ds - \int_t^T Z_s\, dW_s, \tag{2}$$

with ξ a terminal condition; the Z-process is the object that deep BSDE methods learn. Peng [2004] established the link between BSDEs and nonlinear expectations, placing Z in the role of a risk-sensitivity functional. Mean-field games were introduced by Lasry–Lions and Huang–Caines–Malhamé; Carmona and Delarue [2013] and the two-volume monograph of Carmona and Delarue [2018] supply the probabilistic treatment via McKean–Vlasov FBSDEs, and Buckdahn, Li, and Peng [2009b], Buckdahn, Djehiche, Li, and Peng [2009a] establish the MF-BSDE / non-local PDE equivalence and the $1/\sqrt{N}$ propagation of chaos rate that justifies mean-field approximation at finite N. The jump-augmented case has a parallel classical theory [Tang and Li, 1994, Barles, Buckdahn, and Pardoux, 1997] on which any deep BSDEJ method must rest.

Deep BSDE numerics. The curse of dimensionality in classical BSDE schemes [Kloeden and Platen, 1992, Peng and Xu, 2011] — which require conditional-expectation estimation scaling as $O(M^d)$ in state dimension — was broken by the deep BSDE method of Han, Jentzen, and E [2018], which parameterises the gradient Z_t by a neural network and trains on the terminal-loss $|Y_T - \xi|^2$. Beck, E, and Jentzen [2019] extended this to fully nonlinear second-order BSDEs. For mean-field problems, Han et al. [2022] proved curse-of-dimensionality-free convergence for McKean–Vlasov FBSDEs via a fictitious-play decomposition [Guo, Hu, Xu, and Zhang, 2023], with the caveat that the proof requires Lasry–Lions monotonicity on the coupling term — a condition that for the Cont–Xiong kernel (1) can only be verified empirically along a one-dimensional slice of the measure space. The extension to jump-diffusion processes and forward BSDEs with jumps (FBSDEJs) is due to Wang et al. [2023]; the BSDEJ numerical literature is as yet thinly cited, and any downstream application must do its own due diligence on convergence and on the compensated-martingale correction.

Structural pathologies of deep approximations to the MFG HJB system. Germain, Pham, and Warin [2022] catalogue vanishing gradients and terminal-scale sensitivity for the canonical deep BSDE. Three further failure modes are specific to the mean-field and jump-augmented setting and are structural — each corresponds to a specific way in which the neural-network approximation fails to represent the mean-field HJB system it is supposed to solve, rather than a generic optimisation difficulty.

(i) Loss of measurable μ-dependence via BatchNorm. For a mean-field game the value function is $V(t, x, \mu_t)$ and the HJB equation it satisfies is explicitly μ-dependent. When a broadcast population statistic m_t (a low-dimensional surrogate for μ_t) is included as a network input and passed through a BatchNorm layer, normalisation across the batch dimension maps the constant feature to zero, so the parameterised value function becomes measurable with respect to own-state only. The discretised system effectively solves a single-agent HJB, not the MFG.

(ii) Degenerate fixed points of symmetric-input DeepSets. Permutation-invariant pooling of opponent states is the architecturally correct way to encode the empirical distribution. Under symmetric initialisation and symmetric inputs, however, DeepSets admits degenerate minima in which the map

from population state to policy collapses to a constant. The effect is operationally indistinguishable from (i): the approximation again loses measurable dependence on μ.

(iii) Underdetermined terminal loss ("generator bypass"). The deep BSDE terminal loss $|Y_T - g(q_T)|^2$ is a single scalar constraint on the network-parameterised U-process. Many U-profiles satisfy it in expectation, but only one corresponds to the Bellman-optimal $U^\star(q) = V(q-1) - V(q)$ of the underlying HJB system. Without a prior, Adam selects whichever basin is numerically easiest, and the method can converge to a self-consistent fixed point of the BSDE discretisation that does not solve the HJB — the spread profile extracted from such a U is not a best response to the population quote.

Diagnosing and repairing these three pathologies is a core methodological contribution of the project.

Multi-agent RL and tacit collusion. Cont and Xiong [2024] show via decentralised MADDPG that independently learning market-making algorithms converge to spreads strictly above the Nash equilibrium, a phenomenon termed *tacit collusion*. Related MARL work on realistic limit-order-book simulators [Karpe, Fang, Ma, and Wang, 2020, Shi and Cartlidge, 2024, Cheridito, Dupret, and Wu, 2025] highlights fragility across market regimes. The gap between MADDPG spreads and Nash, $|S_{\text{MADDPG}} - S_{\text{Nash}}|$, is a measurable signature of algorithmic collusion, but it has not been operationalised as a detector statistic — which requires a BSDE-based Nash benchmark that is itself trusted. Recent frontier work on common-noise MV-FBSDEs via path signatures [Hu, Jin, and Laurière, 2025] and on mean-field LOB embeddings [Navarro-Rubio and Lage-Castellanos, 2025] defines the open problems beyond this project's scope.

3 Research Question

What are the approximation error and convergence behaviour of a deep BSDE-with-jumps approximation to the N-player Nash equilibrium of the Cont–Xiong mean-field FBSDE, and what structural conditions on the neural parameterisation preserve the mean-field HJB coupling under that approximation? Three sub-questions structure the investigation. (Q1) What is the approximation error $|S_0^{\text{neural}} - S_0^{\text{Nash}}|$, and its scaling in N, for a jump-augmented deep BSDE solver applied to this model; in particular, is the error consistent with the $O(1/\sqrt{N})$ propagation-of-chaos rate of Buckdahn et al. [2009a]? (Q2) Which structural pathologies of the neural parameterisation (§2) admit numerical fixed points that do not solve the underlying HJB system, and which architectural or regularisation choices eliminate them? (Q3) How does the spread gap $|S_{\text{MADDPG}} - S_{\text{Nash}}|$ — a quantitative statistic for deviation of multi-agent RL dynamics from the mean-field Nash equilibrium — behave across model parameters (N, λ, ϕ)?

4 Anticipated Outcomes

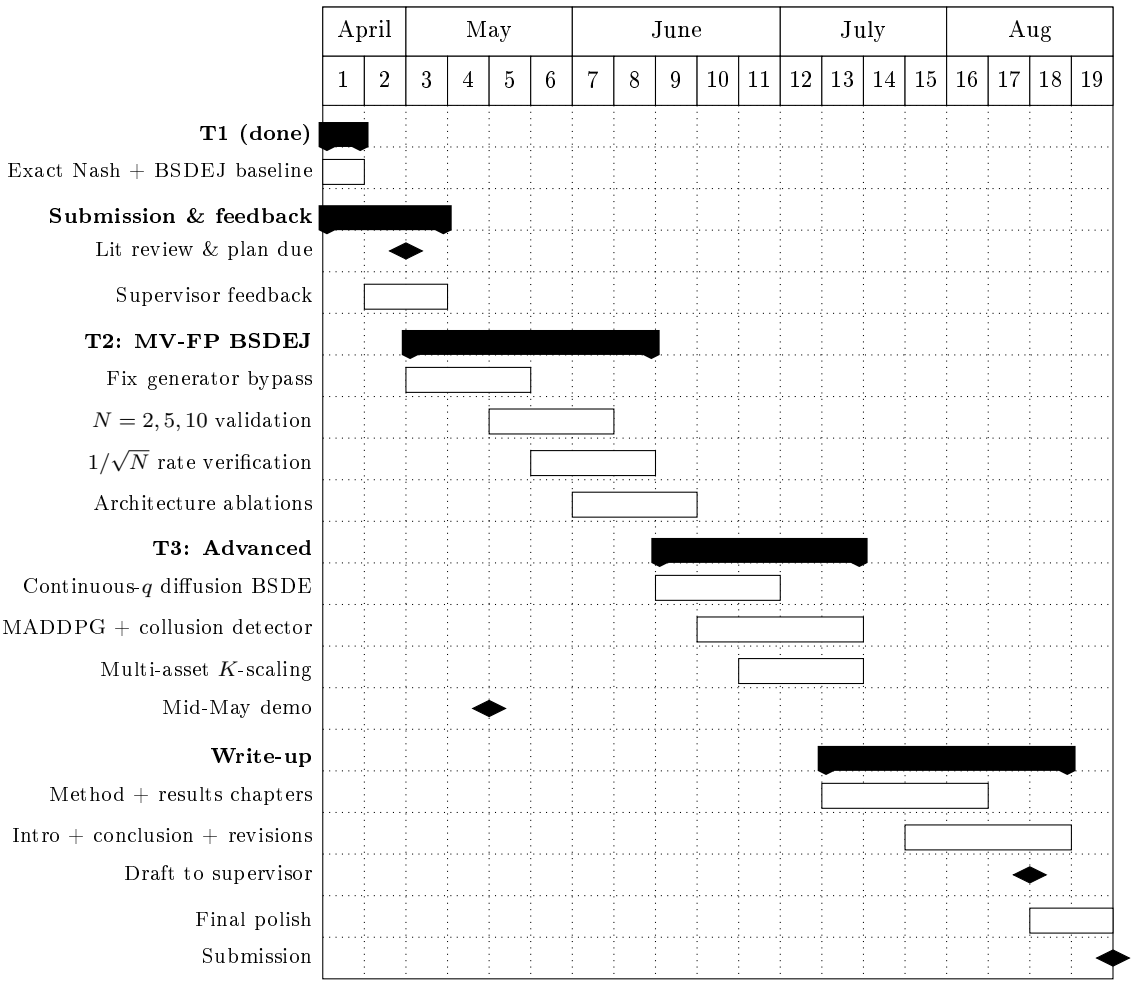
Outcomes are stratified into three levels. Tier 1 (basic) is essentially complete and de-risks the submission. Tier 2 (intermediate) is in progress with known technical challenges. Tier 3 (advanced) is genuinely speculative and is the ceiling, not the floor.

Tier 1 is essentially complete and provides a fall-back position should the more ambitious components prove intractable. Tier 2 contains the principal research content of the project. Tier 3 represents original extensions that may or may not be achievable within the available timeframe; they are reported in the dissertation if completed, and identified as clearly defined open problems if not.

5 Project Time Plan

The project runs from project allocation (January 2026, completed) through submission on 4 September 2026. The core dissertation phase runs 27 April – 26 June 2026, with July–August reserved for write-up. A Gantt summary of the remaining timeline is below.

Tier	Outcome	Risk
T1: Basic	Exact N-player Nash equilibrium of Cont–Xiong via policy-evaluation fictitious play on a discrete inventory grid, validated at $N = 2, 5, 10, 25, 100$ against the Bellman fixed-point tolerance. Shared-weight finite-horizon BSDEJ solver matching Nash spread at $q = 0$ to within 1% with fixed avg_comp.	Low
T1: Basic	Pytest-level unit tests and regression tests for all solvers (boundary-quote convention, compensated-martingale correction, FOC optimality, propagation-of-chaos sanity).	Low
T2: Intermediate	Mean-field fictitious-play BSDEJ solver converging to Nash at $N = 2, 5, 10$ with spread error $< 3\%$. Requires resolving the generator-bypass failure mode identified in §2.	Medium
T2: Intermediate	Empirical verification of $O(1/\sqrt{N})$ propagation of chaos [Buckdahn et al., 2009a] at $N \in \{2, 3, 5, 10, 25, 100, 5000\}$ via exact solver, and tightness of the bound for the neural solver.	Medium
T2: Intermediate	Architecture ablation study (ReLU vs. Tanh vs. GELU; shared vs. per-timestep networks; DeepSets vs. mean-pooling) quantifying each failure mode in §2.	Low
T3: Advanced	Continuous-inventory diffusion BSDE solver with $Z \neq 0$, learning $Z_t = \sigma_q \partial_q V$ genuinely rather than the pure-jump degenerate case.	High
T3: Advanced	MADDPG reproduction of Cont and Xiong [2024] and quantitative operationalisation of the spread gap $\lvert S_{\text{MADDPG}} - S_{\text{Nash}} \rvert$ as a collusion detector across a parameter sweep in (N, λ, ϕ).	High
T3: Advanced	Common-noise extension via path-signature conditioning [Hu et al., 2025]; multi-asset extension ($K = 2, 3, 5$) demonstrating genuine curse-of-dimensionality benefit over the exact solver.	High



Weekly numerals on the chart correspond to one-week periods with week 1 being 27 April 2026 (start of the core dissertation phase). Weekly supervisor meetings are scheduled throughout Term 3; the preliminary demonstration (MiSCADA milestone) is pencilled in for mid-May. The draft handover to the supervisor is scheduled for the end of the third week of July to allow a round of feedback before the September deadline.

6 Demonstrating Success

The project is a numerical one and admits quantitative validation at every tier.

Ground-truth validation. The classical fictitious-play Bellman solver provides an exact reference Nash equilibrium for discrete inventory and small N. All neural solvers are validated against this reference. The principal success metric is the *relative spread error at* $q = 0$,

$$\varepsilon_{\text{spread}} = \frac{|\, S_0^{\text{neural}} - S_0^{\text{Nash}} \,|}{S_0^{\text{Nash}}},$$

with targets of $\varepsilon_{\text{spread}} < 1\%$ at Tier 1 (fixed population quote) and $\varepsilon_{\text{spread}} < 3\%$ at Tier 2 (MV-fictitious-play closure). The current best Tier 1 result is 0.59% error at $N = 2$.

Empirical convergence-rate tests. The $O(1/\sqrt{N})$ propagation-of-chaos bound of Buckdahn et al. [2009a] gives a principled, non-asymptotic check: computing the exact Nash at $N \in \{2, 3, 5, 10, 25, 100, 5000\}$ and fitting $S_0^{\text{Nash}}(N) = a + b/\sqrt{N}$ should yield a residual consistent with the theoretical rate. Deviations flag either a solver bug or a breakdown of the mean-field regime.

Failure-mode diagnostics. Each of the three failure modes identified in §2 admits a *targeted* test. BatchNorm erasure is detected by injecting a known m_t-dependent signal and verifying that the network's output responds to it (controlled-input test). Representation collapse under DeepSets is detected by permuting opponent states and measuring the deviation of the network output (equivariance test). Generator bypass is detected by comparing the network's extracted optimal quotes against the FOC solution to the analytic optimisation problem at each inventory state (FOC-consistency test).

Reproducibility and regression. All solvers are covered by a pytest suite (`tests/test_*.py`) pinned to small problem sizes ($N = 2$, $Q = 5$, $M = 50$ time steps) and numerical tolerances that detect any regression in the compensated-martingale correction or boundary-quote convention. Intermediate results are saved as JSON under `results_final/` for incremental recovery on failure, and all experiments are seeded for deterministic replay.

Demonstrating negative results. The regulations explicitly note that negative results are still results. The project anticipates at least one such outcome: characterising precisely *why* the naive MV-fictitious-play BSDEJ converges to spurious self-consistent fixed points (as observed in early prototyping) is itself a contribution, whether or not the repair succeeds. Partial success is reported transparently, with failure modes documented rather than papered over.

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